

# Jacob Edwards, PhD

## Data Scientist

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portfolio: <https://amnion.github.io/>

Data scientist with 13 years of experience designing experiments and testing models. Specialized in handling time-series signals and other unstructured data. I am action-oriented, excel in ambiguous, fast-paced environments, and approach analytics with disciplined reasoning cultivated in scientific rigor. My goal is to help teams make informed decisions by discerning signal from noise.

### SKILLS

#### Programming & Tools

Python, MySQL, R, MATLAB, MS Office

#### Data Analysis

Statistical modeling, regression and classification, time series analysis, natural language processing, machine learning, feature engineering, dimensionality reduction, model evaluation for explanation and prediction, data visualization

#### Experimentation

Hypothesis testing, experimental design, effect size estimation, uncertainty quantification, bootstrapping, simulation

#### Communication

Data storytelling, technical writing, stakeholder communication, cross-functional collaboration, teaching and mentoring

### RELEVANT EXPERIENCE

#### Research Scientist

COLUMBIA UNIVERSITY

New York, NY

2019 - 2025

Led end-to-end execution of research projects involving large-scale time-series and sequential signal data, from question and hypothesis formulation to experimental design and data collection, through to modeling, validation and delivery to stakeholders.

#### Time-Series Modeling & Workflow Automation

- Generated and analyzed 3.2 TB of stimulus-response time-series neural data. Built scalable workflows for signal cleaning, feature extraction, aggregation and visualization.
- Replaced a manual data quality screening workflow with a supervised ML-based [anomaly detection classifier](#), reducing manual review time by ~20% and improving consistency across projects.
- Implemented statistical and predictive models, including generalized linear models to estimate spectrotemporal receptive fields, enabling the quantitative comparison of neural responses across conditions.
- Built [interactive dashboards](#) in Python (Dash) and MATLAB to support exploratory analysis and enable non-technical collaborators to interpret multi-dimensional analytical results.

#### Sequence Modeling & Structured Data Engineering

- Generated and analyzed large-scale acoustic datasets comprising more than 24,000 vocal events across 138 subjects. Optimized event segmentation pipeline to derive structured tabular representations from raw signals.
- Engineered features from audio recordings using domain metrics and decomposition techniques (PCA, wavelets). Applied dimensionality reduction and clustering techniques to identify latent structure in vocal events.
- Created a [graphical user interface](#) to unify database management and data entry pipelines. The GUI trivialized manual annotation (reduced work hours per project ~80%) and enabled non-technical interns to collaborate on project progress.
- Developed and evaluated NLP model for sequence prediction using non-lexical signal features. Designed a custom cost metric based on sequence distance algorithms. Created custom effect size metrics and bootstrapped confidence intervals.

#### Leadership & Cross-Functional Impact

- Secured \$140,256 in competitive [federal research funding](#) by translating technical proposals into clearly articulated project plans and measurable outcomes.
- Developed and taught an original, [applied data science curriculum](#) (20 hours of lecture, 7 homework assignments) covering statistical modeling, data wrangling, visualization and machine learning. Emphasized the translation of real-world data analysis for policy decision-making and the delivery of actionable insights to stakeholders.
- Authored 6 peer-reviewed publications in high-impact, general and specialist journals including *Nature Scientific Reports* and the *Journal of Neuroscience*.

### SELECTED PUBLICATIONS

|      |                                                                      |                       |
|------|----------------------------------------------------------------------|-----------------------|
| 2026 | Syntax models of species rules accurately predict vocal sequences    | [in press]            |
| 2025 | Temporal organization of behavior is driven by species, not learning | <a href="#">paper</a> |
| 2023 | Machine-learning analysis of song links vocal acoustics to ancestry  | <a href="#">paper</a> |

### EDUCATION

|     |      |                              |
|-----|------|------------------------------|
| PhD | 2025 | Columbia University          |
| MSc | 2018 | Colorado State University    |
| BSc | 2013 | Uni. of Tennessee, Knoxville |